# Lab Guide: Create Virtual Machines

## About this lab

Compute Engine gives you a lot of choice when you build a virtual machine: the machine size, the operating system, the disk, the network settings and how the VM behaves during maintenance. In this lab you build three VMs, each with a different purpose, and look at which settings you can change after creation and which are fixed.

| VM name | Purpose | Key settings |
|---|---|---|
| utility-vm | Small Linux admin machine | E2, e2-medium, no external IP |
| utility-wm | Windows Server machine | E2, e2-standard-2, 64 GB SSD boot disk, HTTP/HTTPS allowed |
| utility-cm | Custom-sized Linux machine | E2 custom, 2 vCPUs, 4 GB memory, Debian |

## What you will learn

- How to create a standard Compute Engine VM and compare machine types and costs
- How to create a Windows VM and prepare it for a Remote Desktop connection
- How to create a VM with a custom machine type and inspect its hardware from the command line

## Before you begin: open the lab environment

Each lab gives you a fresh Google Cloud project, with temporary credentials, for a limited time.

1. Select **Start Lab**. If the lab requires payment, choose a payment method in the pop-up.
2. Look at the **Lab setup and access** panel. You'll find the temporary username and password, the **Open Google Cloud console** button, and the region and zone to use. The countdown timer at the top shows how much time you have left.
3. Select **Open Google Cloud console**. In Chrome, you can right-click it and open the link in an Incognito window instead.
4. When the sign-in page appears, choose **Use Another Account** if you're asked to pick an account.
5. Enter the lab **Username**, select **Next**, then enter the lab **Password** and select **Next**.
6. Accept the terms. Skip recovery options, two-factor authentication and free-trial offers, because this account only exists for the lab.

**Important:** Sign in only with the lab credentials. Using your personal or company Google Cloud account could result in charges.
**Tip:** Keep the lab instructions and the console in separate windows side by side. You can reach any product from the **Navigation menu** at the top left or by typing its name in the search bar.

# Task 1. Build a utility VM

Here you create a small Linux VM of the kind an administrator might use for routine tasks, and explore how pricing changes with machine size.

## Create the VM

1. Go to **Navigation menu** > **Compute Engine** > **VM instances** and select **Create instance**.
2. Name the VM **utility-vm**. The question-mark icon beside the field explains the naming rules.
3. Set **Region** and **Zone** to the values shown in your Lab setup and access panel.
4. Set **Series** to **E2**.
5. Open the **Machine type** list and look through it. Each entry shows the vCPU count, the memory and a short name such as e2-medium. That short name is what you would pass to gcloud on the command line. The monthly cost estimate on the right of the page updates as you choose.
6. Pick **standard** > **e2-standard-4** (4 vCPUs, 16 GB memory) and note how the estimated cost changes.
7. Switch to **Shared-core** > **e2-medium** (2 vCPUs, 4 GB memory). This is the size you'll keep.
8. Open the **Networking** section, expand the network interface, set **External IPv4 address** to **None**, and select **Done**.
9. Keep every other setting at its default and select **Create**. Wait for the VM to finish starting.

**Why no external IP?** An admin VM rarely needs to be reachable from the internet, and external IP addresses outside the Free Tier carry a small charge. See the External IP address pricing section of the VPC documentation for details.

## Look at what you can and can't change

1. On the **VM instances** page, select **utility-vm** to open its details.
2. Find **CPU platform** and write down its value.
3. Select **Edit** and scroll through the page. You'll see that machine type, CPU platform and zone are locked. These are set when the VM is created. Settings such as network tags, firewall options and additional disks can still be changed.
4. Find the boot disk option that controls whether the disk is deleted along with the instance. It's on by default. Turning it off matters when you want to build an image from the boot disk, because you can't create an image from a boot disk while it's attached to a running instance.
5. Select **Cancel** to leave without saving.
6. Review the **Availability policies** section and consider the points in the table below.

| **Setting** | **What to know** |
|---|---|
| Preemptible / Spot | Decided only at creation time. A standard VM can't be switched to preemptible later. Preemptible VMs cost less but can be interrupted at any time. |

| | |
|---|---|
| Automatic restart | Restarts the VM after an outage or hardware failure. Make sure your applications can handle starting up a second time cleanly. |
| On host maintenance | Defaults to live migration. You can choose to have the VM terminated instead. |

**Note:** Some changes, especially network changes such as firewall rules or external IP addresses, can take several minutes to apply.

### Check the VM's logs

1. From the **utility-vm** details page, select **Logging**. This opens a structured log view with filter menus and a search box for labels or text.
2. Expand any log entry using the arrow on its left to see what information it records.

# Task 2. Build a Windows VM

Next you create a Windows Server VM with a larger SSD boot disk and web traffic allowed, then generate a password for it.

### Create the VM

1. Go to **Compute Engine** > **VM instances** and select **Create instance**.
2. Enter the settings below.

| **Setting** | **Value** |
|---|---|
| Name | utility-wm |
| Region and Zone | From your Lab setup and access panel |
| Series | E2 |
| Machine type | standard > e2-standard-2 (2 vCPUs, 8 GB memory) |

3. Open **OS and storage** and select **Change**.
4. Set the Operating **system** to **Windows Server** and choose the version your lab specifies (shown in the lab as windows_image_name).
5. Set **Boot disk type** to **SSD persistent disk** and **Size (GB)** to **64**, then select **Select**.
6. Open **Networking** and, under **Firewall**, turn on **Allow HTTP traffic** and **Allow HTTPS traffic**.
7. Select **Create**.

**Note:** Once the VM is running, the connect button in the last column says **RDP** instead of SSH. Windows VMs are reached through the Remote Desktop Protocol, which needs an RDP client on your computer. Installing one is outside the scope of this lab, so you'll stop just before connecting. The Connecting to Windows VMs guide covers the full process.

## Generate a Windows password

1. Select **utility-wm** to open its details. The VM has no password yet, so nobody can sign in.
2. Select **Set Windows password**, then **Set**.
3. Copy the generated password and select **Close**.

**Note:** In a real deployment, you would now select **RDP** on the VM instances page and sign in with this password using an RDP client, such as one from the Chrome Web Store.

# Task 3. Build a custom-sized VM ( It may not allowed by org Policy - Skip this task 3 in that case)

Predefined machine types don't always match a workload. Here you set your own vCPU and memory values, then confirm them from inside the VM.

## Create the VM

1. Go to **Compute Engine** > **VM instances** and select **Create instance**.
2. Enter the settings below.

| Setting | Value |
|---|---|
| Name | utility-cm |
| Region and Zone | From your Lab setup and access panel |
| Series | E2 |
| Machine type | Custom, with 2 cores and 4 GB memory |

3. Open **OS and storage**. If the image isn't the Debian version your lab specifies (shown in the lab as debian_image_name), select **Change**, choose it, and select **Select**.
4. Select **Create**.

## Inspect the hardware over SSH

1. In the row for **utility-cm**, select **SSH** to open a terminal in your browser.
2. Run each command in the table below and compare the output with the 2 vCPUs and 4 GB of memory you chose.

| Command | What it shows |
|---|---|
| `free` | Used and free memory, plus swap space |
| `sudo dmidecode -t 17` | Details of the installed RAM |
| `nproc` | Number of processors available |
| `lscpu` | Detailed CPU information |
| `exit` | Closes the SSH session |

# Task 4. Review

You built three VMs with different characteristics:

- **utility-vm:** a small shared-core Linux VM with no external IP, used to explore pricing, editable settings, availability policies and logs.
- **utility-wm:** a Windows Server VM with a 64 GB SSD boot disk and web traffic allowed, prepared for RDP with a generated password.
- **utility-cm:** a Debian VM with a custom machine type, whose vCPUs and memory you confirmed from the command line.